THE NOVA RESEARCH CURRICULUM

Computer Science and skills

Grade 7 · Term 3

Student edition

Outcomes: CS-7.3 how data trains AI, bias and limits; CS-7.4 networks, the internet and cybersecurity; RO-7.1; FL-7.1; RP-7.1; AR; PE.

Lesson notes

- **CS-7.3 (Weeks 1–8):** AI learns from labelled examples (training data), then predicts on new data. More varied data usually gives better results. Bias: if the data leaves out some groups, the AI can be unfair (for example, a face filter that works badly on darker skin). Limits: AI can sound confident and still be wrong. Activity: "train" a classmate with picture cards that only show red apples, then test with a green apple. The teacher demonstrates AI tools; no student accounts in Grade 7.
- **CS-7.4 (Weeks 9–12):** LAN and WAN; routers; IP addresses; data travels in packets; Wi-Fi. Cybersecurity: strong passwords (long passphrases), two-step verification, spotting phishing messages, never sharing a one-time code (OTP), looking for HTTPS.
- **RO-7.1:** design brief, build and showcase a sensor-controlled device (for example an automatic night light).
- **FL-7.1:** banks and saving accounts; Islamic banking avoids interest and uses profit-and-loss sharing (Mudarabah, Musharakah) or trade (Murabaha); conventional banks pay and charge interest; the State Bank of Pakistan regulates both.
- **RP-7.1:** write the survey report (question, method, chart, findings, limits) and present it.
- **AR, PE:** Islamic geometric art and a product design; athletics and sports day; health talk on junk food and sugar.

Workbook pages

Page	Task
1	Training-data experiment record.
2	Spot the bias: three case studies.
3	Network diagram; phishing-message checklist.
4	Device design log.
5	Survey report.

Question bank

#	Outcome	Question	Your answer
1	CS-7.3	The examples an AI learns from are called ...	software / training data / hardware / output
2	CS-7.3	If training data leaves out some groups, the AI may be ...	faster / biased / smaller / offline
3	CS-7.3	An AI answer that sounds confident is always correct. True or false.	True / False
4	CS-7.4	A device that sends data between networks is a ...	monitor / router / printer / keyboard
5	CS-7.4	Data on the internet travels in small pieces called ...	bricks / packets / files only / signals only
6	CS-7.4	You should share a one-time code (OTP) with someone who calls from "the bank". True or false.	True / False
7	CS-7.4	Give two signs of a phishing message.	_____
8	FL-7.1	Islamic banking avoids ...	saving / interest (riba) / trade / profit